

Total No. of Questions : 8]

SEAT No. :

PD-4078

[Total No. of Pages : 2

[6402]-38

S.E. (Computer Engineering) (AI & DS) (Computer Science & Design)

COMPUTER GRAPHICS

(2019 Course) (Semester-III) (210244)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3, or Q.4, Q.5, or Q.6, Q.7, or Q.8.
- 2) Draw neat diagram wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Differentiate between Parallel projection and perspective projection. [5]

b) What is transformation and write transformation matrix for: [5]

i) 3D translation using homogenous coordinate system.

ii) 3-D rotation about X-axis.

c) Find transformation of a triangle A(1, 0) B(0, 1) C(1, 1) by performing translation by one unit in x and y directions and then rotating 45° about origin. [8]

OR

Q2) a) Derive 3D transformation matrix for rotation about an arbitrary axis. [5]

b) What are the types of projection and write in brief about any one type of projection? [5]

c) A triangle is defined by [8]

$$\begin{bmatrix} 2 & 4 & 4 \\ 2 & 2 & 4 \end{bmatrix}$$

Find transformed coordinates after the following transformation.

i) 90° rotation about the origin.

ii) Reflection about line $y = -x$

P.T.O.

- Q3)** a) Explain backface detection and removal Algorithm. [6]
b) Explain and compare point source and diffuse illumination. [5]
c) Explain ambient light and diffuse reflection with examples. [6]

OR

- Q4)** a) Write short note on Painters Algorithm. [6]
b) Explain Halftone shading. [5]
c) Write short note on Warnock's Algorithm. [6]

- Q5)** a) Write a short note on interpolation and approximation. [4]
b) Explain Hilbert's curve with an example. [7]
c) Explain blending function for B-spline curve. [7]

OR

- Q6)** a) Explain the Bezier curve. List its properties. [4]
b) Write a short note on Interpolation. [7]
c) With suitable example write short note on the fractal lines. [7]

- Q7)** a) Draw block diagram of NVIDIA workstation and explain it in brief. [7]
b) Define Morphing and write the applications of Morphing. [3]
c) Explain architecture of i860. [7]

OR

- Q8)** a) What are the state-of-the-art Advances in Gaming? [5]
b) Enlist all the steps required to produce real time animation with suitable examples. [6]
c) Why do we need segments? Write the algorithm for changing visibility attribute of segment. [6]

